

Peripherals: Working Document

Digital Audio Player • Rev 11

Author: Alexander Loker. Design decisions, component selection, analysis and verification.

Drafted by: Claude (Anthropic). Structure and prose from the author's direction and source data.

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1. Summary

User-interface and storage peripherals: display and backlight, front and side controls, microSD.

Peripheral	Qty	Part / Status
Display	1	BuyDisplay ER-TFT024IPS-3: 2.4 in IPS, 240 × 320, ST7789V, 4-wire SPI, no touch
Backlight driver	1	ROHM BD1604MUV charge-pump LED driver from SYS, two-level current set by 2N7002-TP (#1-128, #1-130)
Front switches (up, down)	2	C&K; PTS810 SJM 250 SMTR LFS, top-actuated SMD
Rotary encoder with push switch (select)	1	Alps EC12D1564404, 12 mm, vertical, 30 detents / 15 pulses (#1-125)
Side switches (power, volume up, volume down)	3	E-Switch TL1014BF220QG, side-actuated SMD
microSD socket	1	Same Sky MSD-1-A, push-push, card detect (#1-126)
ESD arrays (microSD, encoder)	2	Good-Ark GSER5U007, 4-line, SOT-23-6L (#1-127)

Source: [1], [4], [5], [6], [8], [10], [13], [15], [17].

2. Design Requirements

Requirement	Value
Display	2.4 in IPS, 240 × 320, SPI; ST7789 or ILI9341 driver; glass within ≈ 43 × 60 mm (inside the battery footprint); vendor must publish the FPC pinout and backlight configuration
Front navigation	Two switches (up, down) and one rotary encoder with push switch (select) on the front face
Side controls	Power, volume up and volume down on the enclosure side
Storage	microSD, 1-bit SDMMC, with card detect
Wake from standby	Any input that wakes the player must use an RTC-capable ESP32-S3 GPIO (GPIO0 to GPIO21)
Enclosure	≈ 56 × 75 × 13.2 mm; switch, encoder and socket heights must fit
Constraints	Low noise · Small footprint

3. Interface

Peripheral	Connection
Front and side switches	One GPIO each, active high: switch to +3V3_DIG, 10 kΩ pull-down at the GPIO. DOWN (U0TXD) adds 1 kΩ in series. No hardware debounce; debounced in firmware.
Rotary encoder	A and B: 10 kΩ pull-up to +3V3_DIG at the contact, 10 kΩ series, 10 nF to GND on the GPIO side; decoded by the PCNT peripheral. Common (C) to GND. Push switch: D to +3V3_DIG, E to SELECT (10 kΩ pull-down). ESD array on ENC_A, ENC_B and SELECT; I/O4 unconnected.
microSD	1-bit SDMMC (Section 6): CLK, CMD, DAT0 to GPIOs; DAT1 to DAT3 pulled up only; card detect to a GPIO with a 10 kΩ pull-up; ESD array at the socket.
Display	SPI (SCL with 22 Ω series, MOSI, CS, D/CX) and reset (Section 5). Backlight from the BD1604MUV (Section 7): EN_LED switches it on and off, DIM_LCD selects dim or bright.

Source: [13], [12].

3.1 GPIO Assignment

Signal	GPIO	RTC (wake)	Notes
POWER	GPIO21	Yes	Side switch
VOLUME+ / VOLUME-	GPIO14 / GPIO17	Yes	Side switches
UP / DOWN	GPIO44 (U0RXD) / GPIO43 (U0TXD)	No	DOWN has 1 kΩ series
SELECT	GPIO12	Yes	Encoder push switch
ENC_A / ENC_B	GPIO16 / GPIO15	Yes	Former 32 kHz crystal pins
CLK_SD / CMD_SD / DAT0_SD	GPIO47 / GPIO41 / GPIO42	No	CLK has 22 Ω series at the MCU
CD_SD	GPIO26	No	Low with a card inserted
SCL_LCD / MOSI_LCD / CS_LCD / D/CX_LCD / RESET_LCD	GPIO39 / 40 / 45 / 38 / 18		CS 10 kΩ pull-down holds strap GPIO45 low
EN_LED / DIM_LCD	GPIO48 / GPIO6		EN internal pull-down; gate 100 kΩ pull-down. ADDR_DAC moved to a strap

Source: [12], [13].

4. External Components

Ref	AL-PN	Part	Notes	Qty
J_LCD		Amphenol FCI F32D-1A7Y1-21050	FPC connector for the display: 50 positions, 0.5 mm pitch, right-angle SMT, dual contact (top and bottom), back-flip ZIF, 2.00 mm height, 0.30 mm cable thickness, 30 mating cycles, gold-plated phosphor-bronze contacts, -40 to 105 °C. [3]	1
SW_UP, SW_DN		C&K; PTS810 SJM 250 SMTR LFS	Top-actuated, J-lead SMD; 4.2 × 3.2 mm; 2.5 mm actuator height; 160 gf; 0.15 mm travel; 150,000 cycles; 50 mA at 16 V DC; -40 to 85 °C. The enclosure key presses the actuator. [4]	2
ENC	#1-125	Alps EC12D1564404	Incremental (A and B) with SPST push-on switch; 12 mm, vertical; 30 detents, 15 pulses; actuator length 17.5 mm; detent torque 10 ± 5 mN·m; switch force 6 N, travel 0.5 mm; rated 1 mA at 5 V DC; max / min operating current 10 mA / 1 mA (resistive); 30,000 cycles; -40 to 85 °C. DigiKey 4809-EC12D1564404-ND. [6]	1
SW_PWR, SW_VOLU, SW_VOLD		E-Switch TL1014BF220QG	Side-actuated, right-angle SMD; 4.7 × 3.5 mm; 220 gf; 50 mA at 12 V DC. The enclosure button presses the actuator. [5]	3
LCD		BuyDisplay ER-TFT024IPS-3 (no-touch version)	2.4 in IPS TFT, 240 × 320, ST7789V (COG), 300 cd/m², 80° viewing all directions. Outline with FPC folded 42.72 × 59.46 × 2.3 mm; active area 36.72 × 48.96 mm. 50-pin, 0.5 mm FPC, 0.30 ± 0.03 mm thick. Used in 4-wire 8-bit serial interface I. [1]	1
J_SD	#1-126	Same Sky MSD-1-A	microSD socket, 9 contacts (8 card + card detect); push in, auto eject out; card detect normally open; 1.8 mm above board; 15.05 × 16.15 mm; gold flash. DigiKey 2223-MSD-1-ACT-ND. [8], [9]	1
D_SD, D_ENC	#1-127	Good-Ark GSER5U007	4-line unidirectional ESD array, SOT-23-6L; 5.0 V working; ≈ 0.70 pF; 12 A (8/20 μs). VCC pin to +3V3_DIG. DigiKey 4786-GSER5U007CT-ND. [10]	2
U_LED	#1-128	ROHM BD1604MUV-E2	4-channel charge-pump white LED driver; 1x / 1.5x / 2x; 2.7 to 5.5 V input; 30 mA per channel max (120 mA total); 1.0 MHz typical oscillator; VQFN016V3030. DigiKey BD1604MUV-E2CT-ND. [15], [16]	1
Q_DIM	#1-130	MCC 2N7002-TP	N-channel MOSFET, 60 V, 115 mA, SOT-23. Switches the 32.4 kΩ ISET branch. DigiKey 2N7002-TPMSCT-ND. [17]	1
C (LED driver)	#1-132	Murata GCM188R71E 105KA64D	1 μF, ±10 %, 25 V, X7R, 0603: VBAT, VOUT and both flying capacitors. DigiKey 490-10673-1-ND. [17]	4
R_ISET	#1-131, #1-133	YAGEO RT0402BRD07100KL; RC0402FR-0732K4L	100 kΩ fixed (ISET to GND) and 32.4 kΩ switched (ISET to Q_DIM drain), 0402. [17]	1 + 1

5. Display Interface

FPC pin	Signal	Connection
1	LEDA	Backlight common anode: +LED (BD1604MUV VOUT), 0.1 μF at the connector
2 to 5	LEDK1 to LEDK4	Backlight cathodes, four LEDs in parallel: BD1604MUV LED1 to LED4 current sinks

FPC pin	Signal	Connection
6 to 9	IM0 to IM3	IM3 IM2 IM1 IM0 = 0 1 1 0: 4-wire 8-bit serial interface I (SCL, SDA, D/CX, CSX)
10	RESET	RESET_LCD, GPIO18
11 to 14	VSYNC, HSYNC, DOTCLK, DE	Unused (RGB interface): tied to GND
15 to 32	DB17 to DB0	Unused (parallel bus): tied to GND
33	SDO	Unused output: open
34	SDI (SDA)	MOSI_LCD, GPIO40
35	RD	Unused (parallel read strobe): tied to VDDI
36	WRX (D/CX)	D/CX_LCD, GPIO38
37	D/CX (SCL)	SCL_LCD, GPIO39, 22 Ω series at the MCU
38	CSX	CS_LCD, GPIO45, 10 k Ω pull-down
39	TE	Not connected
40, 41	VDDI	+3V3_DIG, 0.1 μ F (range 1.65 to 3.3 V)
42	VCI	+3V3_DIG (range 2.5 to 3.3 V)
43, 48 to 50	GND	GND
44 to 47	Touch	Not connected (no-touch version)

Serial write clock: 16 ns minimum cycle (62.5 MHz); reads 150 ns minimum (6.7 MHz). Backlight: four LEDs in parallel with a common anode, V_f 3.2 V typical and 3.4 V maximum at 80 mA total (4×20 mA); normal forward current 70 to 80 mA. Logic inputs: $V_{IH} \geq 0.7 VDDI$, $V_{IL} \leq 0.3 VDDI$.

Source: [1], [2] (Sections 2, 3.1, 4.1 to 4.4; ST7789VW Sections 6 and 7.4.3); connections [13].

6. microSD Interface

Card pin	Signal	Connection
1	DAT2	10 k Ω pull-up to +3V3_DIG
2	CD / DAT3	10 k Ω pull-up (keeps the card in SD mode at power-up); ESD array I/O1
3	CMD	CMD_SD, GPIO41; 10 k Ω pull-up; ESD array I/O2
4	VDD	+3V3_DIG; 10 μ F + 0.1 μ F at the socket
5	CLK	CLK_SD, GPIO47; 22 Ω series at the MCU; no pull-up; ESD array I/O4
6	VSS	GND
7	DAT0	DAT0_SD, GPIO42; 10 k Ω pull-up; ESD array I/O3
8	DAT1	10 k Ω pull-up to +3V3_DIG
CD switch	CD_SD	GPIO26; 10 k Ω pull-up; reads low with a card inserted
G1 to G4	Shell	GND

Source: [13], [8], [11]. Card pin numbering per the standard microSD pinout; confirm against the MSD-1-A drawing [9].

7. Backlight Driver

Pin	Name	Connection
7	VBAT	SYS; 1 μ F X7R at the pin
1	EN	EN_LED (GPIO). Internal pull-down ($\approx 18 \mu$ A at 5.5 V): backlight off while the GPIO floats at reset
2, 3, 4	SEL0, SEL1, SEL2	SEL0 10 k Ω to GND; SEL1 and SEL2 100 k Ω to VBAT. 1-1-0 = LED1 to LED4 all on
5	ISET	100 k Ω to GND (fixed); 32.4 k Ω to the 2N7002 drain (switched). Gate on DIM_LCD with 100 k Ω to GND; source to GND
8, 9 / 10, 11	C1P, C1N / C2N, C2P	1 μ F X7R flying capacitors
6	VOUT	+LED to FPC pin 1 (LEDA); 1 μ F X7R at the pin
13 to 16	LED1 to LED4	FPC pins 2 to 5 (LEDK1 to LEDK4)
12, 17	GND, EPAD	GND

Dimming is set by switching ISET resistance rather than PWM on EN: the datasheet requires a 100 to 200 Hz PWM period at duty cycles of 10 % or less, which would put 80 mA steps in the audio band on SYS and in the ground return. Switched ISET keeps the LED current DC at both levels.

Source: [16], [13].

8. Equations and Results

Quantity	Expression	Result	Source
Encoder edge, rising (contact opens)	$\tau = (10 \text{ k}\Omega + 10 \text{ k}\Omega) \times 10 \text{ nF}$; $t = \tau \ln 4$ to $0.75 \times 3.3 \text{ V}$	0.20 ms; 0.28 ms	Calc.
Encoder edge, falling (contact closes)	$\tau = 10 \text{ k}\Omega \times 10 \text{ nF}$; $t = \tau \ln 4$ to $0.25 \times 3.3 \text{ V}$	0.10 ms; 0.14 ms	Calc.
Quadrature edge spacing at 5 rev/s	$1 / (15 \text{ pulses} \times 4 \text{ edges} \times 5 \text{ rev/s})$	3.3 ms	Calc., [6]
Encoder standby current	$3.3 \text{ V} / 10 \text{ k}\Omega$ per closed contact (A defined at detents)	0.33 mA	Calc., [7]
Select switch current	$3.3 \text{ V} / 10 \text{ k}\Omega$	0.33 mA (vs 1 mA min listed)	Calc., [6]
Backlight current, dim (Q_DIM off)	$I_{\text{LED}} = 480 / R_{\text{SET}}$; $R_{\text{SET}} = 100 \text{ k}\Omega$	4.8 mA per LED; 19 mA total	[16]
Backlight current, bright (Q_DIM on)	$R_{\text{SET}} = 100 \text{ k}\Omega$ 32.4 k Ω = 24.5 k Ω	19.6 mA per LED; 78 mA total	[16]
Backlight current tolerance	ISET voltage 0.5 / 0.6 / 0.7 V	$\approx \pm 17 \%$	[16]
Q_DIM branch current	$0.6 \text{ V} / 32.4 \text{ k}\Omega$	$\approx 19 \mu\text{A}$	Calc.
Charge-pump switching	Oscillator, 1.5x / 2x modes only	0.8 / 1.0 / 1.2 MHz	[16]
LED pin headroom (1x mode limit)	V_{LED} minimum at LED1 to LED4	0.08 / 0.10 / 0.20 V	[16]
Backlight on +3V9 (rejected)	253 mA analog peak + 80 mA	$\approx 333 \text{ mA}$ vs 250 mA design, 0.3 A ceiling	[14]

9. Open Problems

#	Problem
1	Backlight layout: driver GND and LED return straight to power ground, away from the analog region; flying and VOUT capacitors close to the pins.
2	Display supply: VCI and VDDI allow up to 3.3 V. +3V3_DIG is 3.318 V nominal, 18 mV above the recommended maximum and far below the 4.6 V absolute maximum. Accept it or lower the digital rail.
3	Encoder push switch: Alps lists 10 mA / 1 mA max / min operating current (resistive). If the minimum applies to the switch, the 10 kΩ SELECT pull-down (0.33 mA) is below it; ≈ 3.3 kΩ meets 1 mA. Confirm in the EC12 catalog.
4	Encoder standby current ≈ 0.33 mA whenever it rests on a detent with A closed. Accept, or raise the pull-ups (rescale the RC).
5	Encoder fit: 17.5 mm actuator length against the 13.2 mm enclosure; knob height and protrusion to be modelled.
6	microSD socket: confirm in the MSD-1-A drawing that the card-detect switch closes to the shell; position the socket for the enclosure slot and push-push eject travel.
7	Housekeeping: replace 0603 10 kΩ with the 0402 part (#1-134); library rows #1-131 (tolerance column reads 1 %, part is 0.1 %) and #1-134 (size column reads 402).
8	Optional: 1 kΩ series on SELECT, VOLUME+ and VOLUME- against the power-up glitch (MCU Rev 9, C17).

Source: [1], [6], [7], [9], [12], [13], [16], [17].

10. Revision Log

Rev	Date	Change
0	2026-09-27	Initial issue: peripheral set, requirements, interface.
1	2026-09-27	Side switches: E-Switch TL1014BF220QG.
2	2026-09-27	TS126CJ specifications recorded; encoder changed to Grayhill 62V11-02-P.
3	2026-09-27	Front switches changed to three C&K; PTS810 (up, down, select); encoder changed to Alps EC10E1220501.
4	2026-09-27	Display selected: BuyDisplay ER-TFT024IPS-3 (ST7789V, 4-wire SPI).
5	2026-09-27	Display connector: Amphenol FCI F32D-1A7Y1-21050.
6	2026-09-27	F32D specifications confirmed from the manufacturer description: 2.00 mm height, 0.30 mm cable, 30 mating cycles.
7	2026-09-27	Panel datasheet values recorded: outline, FPC thickness, interface mode, supplies, backlight. Backlight drive and supply-level notes added.
8	2026-09-27	Serial data pin and unused-pin terminations from the ST7789VW datasheet; SPI clock limits recorded.
9	2026-10-01	Encoder changed to Alps EC12D1564404 with push switch as SELECT; front switches reduced to two PTS810. Buttons active high (pull-down, switch to +3V3_DIG), DOWN 1 kΩ series, firmware debounce. microSD added (Same Sky MSD-1-A, 1-bit SDMMC). ESD arrays GSER5U007 on microSD and encoder. GPIO assignment, display pin connections, microSD interface and results recorded. Closed: encoder RC values, GPIO assignment. Added: SELECT tied to +3V3_DIG, encoder switch current, standby current, encoder fit, socket checks.

Rev	Date	Change
10	2026-10-02	Backlight driver added: ROHM BD1604MUV from SYS, all four channels on via SEL straps, two-level current by switched ISET (100 kΩ fixed, 32.4 kΩ via 2N7002-TP): 4.8 / 19.6 mA per LED. Driver capacitors 1 μF X7R 0603. Display LEDA / LEDK connections updated. Closed: backlight drive selection, SELECT tied to +3V3_DIG. Added: ISET fixed resistor check, backlight GPIOs, backlight layout, housekeeping.
11	2026-10-02	ISET fixed 100 kΩ placed. Backlight pins assigned: EN_LED GPIO48, DIM_LCD GPIO6. Closed: ISET fixed resistor, backlight GPIOs.

11. Sources

1. ER-TFT024IPS-3 Datasheet (EastRising / BuyDisplay), project file
2. ST7789VW Controller Datasheet v1.0 (Sitronix), project file
3. [F32D-1A7Y1-21050 Product Page \(DigiKey\)](#)
4. PTS810 SJM 250 SMTR LFS Product Page and Datasheet (DigiKey); link carried from Rev 8 not available
5. TL1014BF220QG Product Page and Datasheet (DigiKey); link carried from Rev 8 not available
6. [EC12D1564404 Product Page \(Alps Alpine\)](#)
7. [EC12 Series Catalog Pages \(Alps Alpine, via Farnell\)](#)
8. [MSD-1-A Product Page \(Same Sky\)](#)
9. [MSD-1-A Datasheet \(Same Sky\)](#)
10. [ESD Transient Voltage Suppressors, GSER5U007 \(Good-Ark Semiconductor\)](#)
11. [ESP32-S3 Hardware Design Guidelines, Schematic Checklist \(Espressif\)](#)
12. MCU (ESP32-S3): Working Document, Rev 9
13. DAP schematic sheet DAP-Peripherals-4 (author, 2026-10-02)
14. Power Budget Rev 6 and Analog Buck-Boost (LTC3440) Rev 7, project files
15. [BD1604MUV Product Page \(ROHM\)](#)
16. [BD1604MUV / BD1604MVV Datasheet \(ROHM, via Mouser\)](#)
17. Author library rows #1-128 to #1-134 (DigiKey part data: BD1604MUV-E2, 2N7002-TP, RT0402BRD07100KL, GCM188R71E105KA64D, RC0402FR-0732K4L, AC0402FR-0710KL)